

```
print(Switch is on?', self.switchIsOn)
print('Brightness is:', self.brightness)
```

Listing 2-5: The slightly more complicated `DimmerSwitch` class

In this `__init__()` method we have two instance variables: the familiar `self.switchIsOn` and a new one, `self.brightness`, which remembers the brightness level. We assign starting values to both instance variables. All other methods can access the current value of each of these. In addition to `turnOn()` and `turnOff()`, we include two new methods for this class: `raiseLevel()` and `lowerLevel()`, which do exactly what their names imply. The `show()` method is used during development and debugging and just prints the current values of the instance variables.

The main code in [Listing 2-6](#) tests our class by creating a `DimmerSwitch` object (`oDimmer`), then calling the various methods.

File: `OO_DimmerSwitch_with_Test_Code.py`

```
# DimmerSwitch class with test code

class DimmerSwitch():
    --- snipped code of DimmerSwitch class, as in Listing 2-5 ---

# Main code
oDimmer = DimmerSwitch()

# Turn switch on, and raise the level 5 times
oDimmer.turnOn()
oDimmer.raiseLevel()
oDimmer.raiseLevel()
oDimmer.raiseLevel()
oDimmer.raiseLevel()
oDimmer.raiseLevel()
oDimmer.show()

# Lower the level 2 times, and turn switch off
oDimmer.lowerLevel()
oDimmer.lowerLevel()
oDimmer.turnOff()
oDimmer.show()
```